What AI Forecasters Know, Use, and Update

Cristina Fuentes-Albero, Manuel González-Astudillo, and Michael Siemer

Discussion by Minchul Shin[1] (FRB Philadelphia)

System Committee meeting
November 5, 2026

[1] **Disclaimer:** The views expressed here are my own and do not necessarily represent the views of the Federal Reserve Bank of Philadelphia or the Federal Reserve System.

This paper . . .

Studies LLM macroeconomic forecasts and their response to new information.

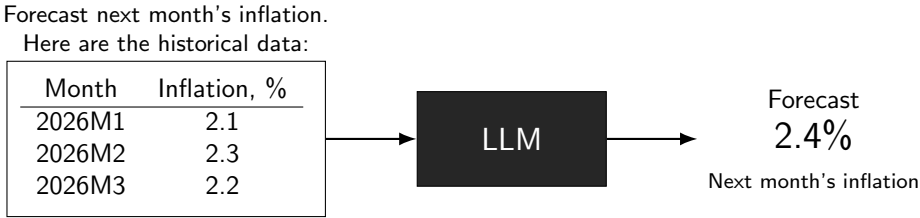
Real data

- Forecast 24 U.S. variables from 2001 to 2024.
- Vary prompts, including forecast versus recall tasks and instructions to ignore later information.
- Find that instructions do not fully remove recall from historical forecasts.

Simulated data

- Forecast six series generated by a VAR.
- Vary prompts, including histories, labels, release reliability, and information about the data-generating process.
- Find that forecasts and updates can depart from MSE-optimal benchmarks.

Comment 1: LLM as a black-box forecasting tool

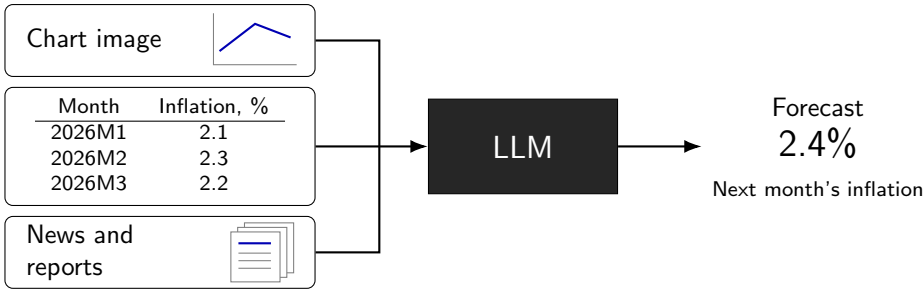


The paper varies the inputs to the LLM

- History length, including 12 versus 60 observations in the simulated data.
- Variable labels and prompt instructions.
- Information about the data-generating process.

This paper explores a limited part of the context an LLM could use.

Comment 1: LLM as a black-box forecasting tool



This paper explores a limited part of the context an LLM could use.

Frontier LLMs support multimodal inputs and million-token context windows.

- Which information, in what form, and how much should an LLM receive?
- For example, books and research papers on forecasting, chart images could be supplied.

Comment 2: LLM as a forecaster

What does ChatGPT do when asked to forecast with tools available?

- I used ChatGPT 6-Pro in a temporary chat (ChatGPT App, November 1, 2026)
- ChatGPT 6-Pro comes with its own cloud computer and several out-of-box tools like internet search, Python execution, etc.

My prompt (intentionally vague)

Here is a time-series of inflation rate of a country. Generate forecasts for the next three months:
{a table of 57 monthly observations, January 2022 to September 2026, CPI inflation from Japan.}

Temporary chat

ChatGPT

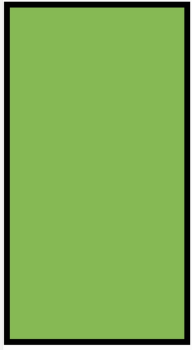
New chat

Your dot 1

Pinned



Projects



Recents

Here is a time-series of inflation rate of a country. Generate forecasts for the next three months

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual
2026	1.5%	1.3%	1.4%	1.3%	1.5%	1.7%	2.1%	2.0%	2.2%	3.0%	2.9%	2.1%	3.2%
2025	4.0%	3.6%	3.6%	3.5%	3.4%	3.2%	3.0%	2.7%	2.8%	2.2%	2.9%	3.7%	2.7%
2024	2.1%	2.8%	2.7%	2.5%	2.9%	2.9%	2.7%	3.0%	2.5%	3.3%	2.9%	2.6%	3.3%
2023	4.4%	3.3%	3.3%	3.5%	3.2%	3.3%	3.3%	3.1%	3.0%	3.8%	3.8%	4.0%	2.5%
2022	0.5%	0.9%	1.2%	2.4%	2.4%	2.3%	2.6%	3.0%	3.0%	3.8%	4.0%	2.5%	

Worked for 1m 25s

Analyzed

Python

```
import numpy as np
import pandas as pd
from scipy.stats import norm
from statsmodels.tsa.holtwinters import ExponentialSmoothing, SimpleExpSmoothing
import warnings

values = [
    0.5,0.9,1.2,2.4,2.4,2.3,2.6,3.0,3.0,3.8,3.8,4.0,
    4.4,3.3,3.3,3.5,3.2,3.3,3.3,3.1,3.0,3.3,2.9,2.6,
    2.1,2.8,2.7,2.5,2.9,2.9,2.7,3.0,2.5,2.2,2.9,3.7,
    4.0,3.6,3.6,3.5,3.4,3.2,3.0,2.7,2.8,3.0,2.9,2.1,
    1.5,1.3,1.4,1.3,1.5,1.7,2.1,2.0,2.2
]
y = pd.Series(values, index=pd.date_range("2022-01-01", periods=len(values), freq="MS"), name="Infl
print(len(y), y.index[0], y.index[-1])
```

Ask ChatGPT

Pro

Temporary chat

These forecasts retain the definition of inflation used in your table; they are not forecasts of cumulative price growth over the next three months.

How I obtained these forecasts

I used the **57 monthly observations from January 2022 through September 2026**, excluding the "Annual" column. I compared eight simple forecasting methods using an expanding-window backtest: starting with 24 observations, each model forecast the next three months, then the estimation window advanced by one month. This produced **31 forecast origins**, with the same evaluation dates for every model.

The best-performing methods were very close:

Model	Backtest RMSE, pooled across the three horizons
Last observed rate	**0.555 percentage points**
Simple exponential smoothing	0.556 percentage points
Damped-trend exponential smoothing	0.557 percentage points
AR(1), with intercept	0.559 percentage points
AR(2), with intercept	0.575 percentage points

Given these results, I selected the simplest model: **forecast the latest observed rate, 2.2%, at each horizon.** Its narrow backtest advantage should not be interpreted as statistically decisive.

What about the recent upward movement?

Inflation increased from 1.3% in April to 2.2% in September. An AR(1) model fitted to the full sample forecasts **2.3%, 2.4%, and 2.5%** for October, November, and December, respectively. That provides a reasonable alternative trajectory, but it did not outperform the flat forecast in the historical evaluation.

The prediction intervals above use the horizon-specific backtest errors and a normal approximation. With a short sample and overlapping forecast errors, their coverage is approximate. **The flat point forecast represents a conservative central estimate—not a prediction that inflation will literally remain unchanged.**

ChatGPT can make mistakes. Check important info.

Ask ChatGPT

Pro

ChatGPT compares models and selects a forecast

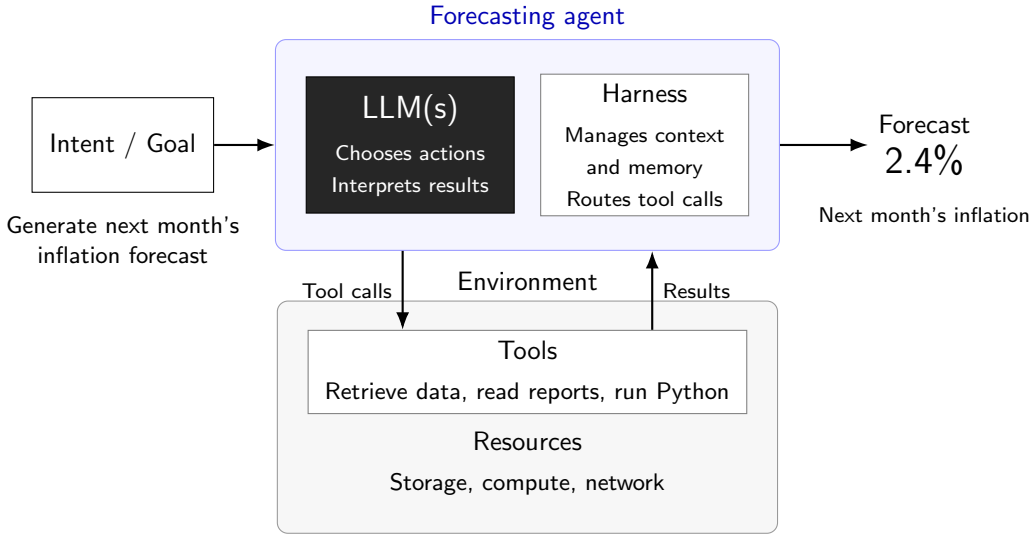
Starts with 24 observations and expands the window one month at a time. Forecasts the next three months at each of 31 origins.

Reported method	RMSE, percentage points
Last observed rate (RW)	0.555
Simple exponential smoothing	0.556
Damped-trend exponential smoothing	0.557
AR(1), with intercept	0.559
AR(2), with intercept	0.575

RMSE pools errors across the three forecast horizons.

> "Given these results, I selected the simplest model: **forecast the latest observed rate, 2.2%, at each horizon.** Its narrow backtest advantage should not be interpreted as statistically decisive."

Comment 2: LLM as a forecaster



Macro and financial forecasting with AI agents

How do these AI systems perform in economic and financial forecasting?

Emmanoulopoulos et al., 2025	Builds stochastic models of stock prices and market risk using real and simulated series.
Ang et al., 2025	TS-Agent selects and tunes models for stock prices, exchange rates, and cryptocurrencies.
Li et al., 2025	R&D-Agent-Quant develops factors and models to predict stock returns through backtests.
Zhao et al., 2025	TimeSeriesScientist selects and combines models for electricity, weather, exchange rates, and influenza.
Shin, 2026	Searches forecast-combination rules for euro area GDP growth, with a separate holdout test.
Zhao et al., 2026	LiveMacroEval uses web-enabled agents to nowcast 16 U.S. indicators, including GDP, inflation, and employment.

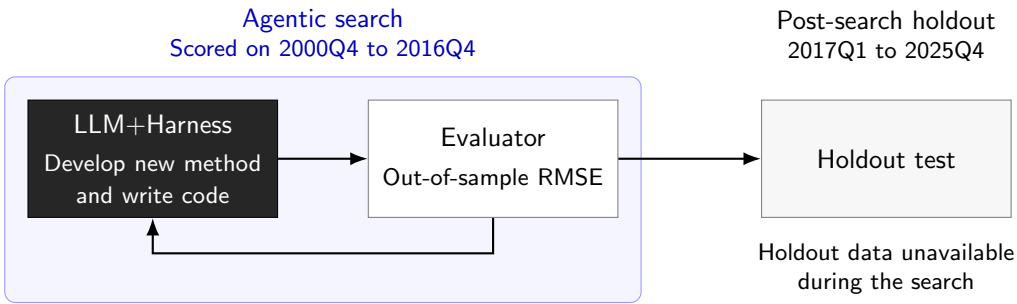
There is also a broader literature on agents that conduct analysis, run experiments, and develop new methods.

Nam et al., 2026a	DS-STAR automates data analysis, with prediction tasks for sales, electricity prices, and food demand.
Chen et al., 2026	MARS searches and refines machine-learning code under compute budgets, evaluated on 75 Kaggle competitions.
Aygün et al., 2026	ERA searches over scientific code, with applications to genomics and forecasts of COVID hospitalizations.
Gottweis et al., 2026	AI co-scientist generates and refines biomedical hypotheses, including drug repurposing and antimicrobial resistance.
Meng et al., 2026	ScientistOne develops algorithms and writes papers, tracing claims to code, experiment logs, and sources.
Schmidgall et al., 2026	Extends AI co-scientist to experimental workflows in materials science, biology, and medical AI.
Nam et al., 2026b	ScientistTwo develops and tests methods, including forecasts of electricity, weather, and exchange rates.

Shin (2026) Agentic search for forecast combination

Combine 23 professional forecasts of euro area GDP growth.

Apply agentic search to find a better forecast combination method.



Shin, M. (2026). "An auditable AI agent loop for empirical economics: A case study in forecast combination." *Economics Letters*, 268, 113164.

Conclusions

Comment 1: LLM as a forecasting tool

- Large context windows and multimodal inputs allow much richer context designs.
- Which information, in what form, and how much should an LLM receive?
- How sensitive are the paper's findings to these choices?

Comment 2: LLM as a forecaster

We need to understand their decisions.

- Which information and forecasting methods do they choose?
- Do they research enough before constructing an empirical model?
- Do recursive research loops improve forecasts enough to justify the added time and cost?